AVERAGE AND EXTREMAL POWER-FLOW EQUATIONS

Abstract. We use probability and reinforcement learning (hopefully) to understand average and extremal behaviour of power-flow equations.

1. Introduction

Power-flow equations correspond to the following collection of equations (citations here):

$$\|A_1x\|^2 = \|A_2x\|^2 = \cdots = \|A_nx\|^2 = 1$$

where A_i are $n \times n$ matrices and $x \in \mathbb{R}^n$. We are interested in understanding how many solutions this system of equations can have, and the landscape of tuples of matrices $(A_1, A_2, \ldots, A_n)$ corresponding to the average and extremal behavior. Algebraic geometry results tell us that there are at most 2^n many solutions over complex numbers if the number of solutions is finitely many (cite affine Bezout exposition of Tao). However, it is unclear what are the possible number of solutions over the reals (cite $\binom{2n}{n-1}$ type result and conjectures). The number real solutions to power-flow equations is important for stability analysis of corresponding networks (cite), but the number of complex solutions does not seem to bear any practical insight. Therefore, in this sense, the important part of the picture is missing. In this paper we aim to shed light on this using probabilistic tools and reinforcement learning.

Our scheme is as follows: We will first derive an estimate on the expected number of solutions when $A_1, A_2, \ldots, A_n$ are random Gaussian matrices with every entry independent, centered at zero, and have the same variance. The expected number of solutions turns out to be $e^{\frac{n+2}{2}}$ up to constant, and we regard this as the average-case behavior. Our second step is to derive a formula for the expected number of solutions for $A_i = B_i + \delta X_i$ for each $i = 1, 2, \ldots, n$ where B_i are deterministic matrices, $\delta > 0$ is the magnitude of our perturbation, and X_i random Gaussian matrices with independent entries centered at zero and variance one. This formula is a proxy to the exact number of real solutions to the system equations

$$\|B_1x\|^2 = \|B_2x\|^2 = \cdots = \|B_nx\|^2 = 1.$$

One might wonder why we prefer a proxy instead of performing an exact count: The reason is that current state of the art methods from computational algebra for exact counting turn computationally intractable before n can become a three digit integer. Our last mathematical step is to use the proxy formula for creating a value function for the 'state' $(B_1, B_2, \ldots, B_n)$. We will use this value function for designing policy-gradient algorithm that will search for extremal systems with maximal 'value'. That is, the policy-gradient algorithm will search in the space of equations for the ones that have approximately the most number of solutions. Once these 'most-value' systems are located, our proxy can be refined for a better approximation to the exact zero count and this will be discussed along the way.

2. Probabilistic Setup and Rice Formula

We prefer a strange beginning and a happy ending. So, we start with stating our technical tool which is Theorem 6.7 in Level Sets and Extrema of Random Fields and Processes by Azais and Wschebor `https://www.math.univ-toulouse.fr/~azais/styles/other/student/level.pdf`, published version is by Wiley. The result requires some assumptions:

(1) We have random field $Z(t) \in \mathbb{R}^d$ that is created as $Z(t) = H[Y(t)]$ where

$$\{Y(t) : t \in W\}$$

is a Gaussian field indexed by a set $W \in \mathbb{R}^d$ and having values $Y(t) \in \mathbb{R}^n$ for some $n \geq d$, and $H : \mathbb{R}^n \to \mathbb{R}^d$ is a continuously differentiable function.

(2) $Z(t)$ has a density $p_{Z(t)}(x)$ that is continuous function of the pair $(t, x) \in W \times \mathbb{R}^d$

(3) $\mathbb{P}\{\exists t \in W : Z(t) = u \ , \ \det(Z'(t)) = 0\} = 0$

Then, we have

$$\mathbb{E}\left(N_u(Z, B)\right) = \int_B \mathbb{E}\left(|det(Z'(t))| \mid Z(t) = u\right) \ p_{Z(t)}(u) \ dt \tag{1}$$

where $N_u(Z, B)$ denotes the number of points in B that satisfies $Z(t) = u$, and we assume B is a Borel set.

2.1. **Random Matrices with Entries Centered at Zero.** For $A_1, A_2, \ldots, A_n$ independent $n \times n$ random matrices that have independent Gaussian entries that are centered at 0 and standard deviation σ, we are interested in counting the expected number of real solutions to the following collection of equations:

$$\|A_1 x\|^2 = \|A_2 x\|^2 = \cdots = \|A_n x\|^2 = 1. \tag{2}$$

So, our random fields is $Z(x) = H[Y(x)]$ is indexed in $W = \mathbb{R}^n$, and $Y(x)$ is given by

$$Y : \mathbb{R}^n \to \mathbb{R}^{n^2} \ , \ Y(x) = (A_1 x, A_2 x, \ldots, A_n x)$$

Then our map H on $y = (y_1, y_2, \ldots, y_n) \in \mathbb{R}^{n^2}$ where $y_i \in \mathbb{R}^n$ for all $i = 1, 2, 3, \ldots, n$ is defined as follows:

$$H : \mathbb{R}^{n^2} \to \mathbb{R}^n \ , \ H(y_1, y_2, \ldots, y_n) = (\|y_1\|^2, \ldots, \|y_n\|^2)$$

where $\|.\|$ denotes the standard Euclidean norm. Using the Rice formula provided by Azais and Wschebor in Equation (1) for $u = (1, 1, 1, \ldots, 1)$ we have

$$\mathbb{E}\left(N_u(Z, \mathbb{R}^n)\right) = \int_{\mathbb{R}^n} \mathbb{E}\left(|det(Z'(x))| \mid Z(x) = u\right) \ p_{Z(x)}(u) \ dx \tag{3}$$

Now we need to unravel the density $p_{Z(x)}(u)$:

$$p_{Z(x)}(1, 1, \ldots, 1) = p_{H(Y(x))}(1, 1, \ldots, 1) = P_{H(Y(\|x\|e_1))}(1, 1, \ldots, 1)$$

Where the second equality follows from rotational invariance of the Gaussian field Y. Now note that $A_1, A_2, \ldots, A_n$ are independent, so we have

$$P_{H(Y(\|x\|e_1))}(1, 1, \ldots, 1) = P_{H(Y\|x\|e_1)}(1, \ldots, 1) = \left(P_{\|x\|^2\|A_1 e_1\|^2}(1)\right)^n$$

The random variable $\|x\|^2\|A_1 e_1\|^2 = \|x\|^2 \left(\psi_1^2 + \ldots + \psi_n^2\right)$ where $\psi_i \sim \mathcal{N}(0, \sigma^2)$, and $\frac{\|x\|^2\|A_1 e_1\|^2}{\|x\|^2\sigma^2}$ follows a χ_n^2 distribution. So we have

$$P_{\|x\|^2\|A_1 e_1\|^2}(1) = P_{\chi_n^2}\left(\frac{1}{\sigma^2\|x\|^2}\right)\frac{1}{\|x\|^2\sigma^2}$$

and

$$p_{Z(x)}(1, 1, \ldots, 1) = \left(\frac{1}{2^{\frac{n}{2}}\Gamma(\frac{n}{2})}\left(\frac{1}{\sigma^2\|x\|^2}\right)^{\frac{n}{2}-1} e^{-\frac{1}{2\sigma^2\|x\|^2}} \frac{1}{\|x\|^2\sigma^2}\right)^n. \tag{4}$$

2.2. **Random Matrices with Entries Centered Arbitrarily.** We will restrict our attention to the following model of randomness:

$$A_i = B_i + \delta X_i \text{ with } \|B_i\|_F = 1 \text{ , } B_1^T B_1 + \ldots + B_n^T B_n = \mathbb{I},$$

where B_i are deterministic matrices and X_i are random Gaussian matrices with independent entries of variance one and expectation zero. Here $\|.\|_F$ is the norm introduced by the following inner product on matrices:

$$\langle X, Y \rangle := \text{Trace}(Y^T X).$$

That is, we have $\text{Trace}(B_i^T B_i) = 1$ for each $i = 1, 2, 3, \ldots, n$. We recall the following basic fact to motivate our choice:

Lemma 2.1 (Barvinok Partition Functions book Thm 4.5.9)**.** *Let $M_1, M_2, \ldots, M_n$ be positive definite matrices, then there exists an invertible matrix Z such that*

$$\text{Trace}(Z^T M_i Z) = 1 \textit{ for each } i = 1, 2, \ldots, n, \textit{ and } M_1 + M_2 + \ldots + M_n = \mathbb{I}$$

This lemma implies the collection of deterministic centers in our model recovers the entire parameter space: An invertible change of variables does not change the number of solutions to the systems of equations

$$\|B_1 x\|^2 = \|B_2 x\|^2 = \ldots = \|B_n x\|^2 = 1 \text{ for } i = 1, 2, \ldots, n.$$

Our aim is to use the expected number of real zeros for the random system $(A_1, A_2, \ldots, A_n)$ as a proxy to counting the number of real zeros of the system $(B_1, B_2, \ldots, B_n)$.

Now suppose we have $x \in \mathbb{R}^n$ such that $\|A_i x\|^2 = 1$ for all $i = 1, 2, \ldots, n$. Then we have

$$n = \sum_{i=1}^{n} x^T A_i^T A_i x = \|x\|^2 + x^T \left(\delta \sum_{i=1}^{n} (X_i^T B_i + B_i^T X_i) + \delta^2 \sum_{i=1}^{n} X_i^T X_i \right) x$$

We note that

$$X := \delta \sum_{i=1}^{n} (X_i^T B_i + B_i^T X_i) + \delta^2 \sum_{i=1}^{n} X_i^T X_i$$

is a real symmetric matrix, and we have

$$n = \|x\|^2 + x^T X x.$$

Moreover, $X \sim \delta^2 n \, \mathbb{I}$ with high probability. Therefore, with high probability, we have

$$n \sim (1 + \delta^2 n) \|x\|^2.$$

Following our notation in Section 2.1, we define the maps Y, H and Z the same way, and the same formula Equation (3) holds true. We note the following fact:

$$\mathbb{E}\left(|\det Z'(x)| | Z(x) = (1, 1, \ldots, 1) \right) p_{Z(x)}(1, 1, \ldots, 1) = \int_{K(x)} |\det Z'(x)| \, d\mu(X_1, X_2, \ldots, X_n)$$

where μ is the induced product Gaussian measure on $(X_1, X_2, \ldots, X_n)$, and

$$K(x) := \{(X_1, X_2, \ldots, X_n) : \|A_1 x\|^2 = \|A_2 x\|^2 = \cdots = \|A_n x\|^2 = 1\}.$$

3. Expected Number of Common Intersections for Random Matrices with Entries Centered at Zero

First note that $Z(x) = (\|A_1x\|^2, \|A_2x\|^2, \dots, \|A_nx\|^2)$ and we need to compute $Z'(x)$. For convenience let us define the following map that sends $x \in \mathbb{R}^n$ to an $n \times n$ matrix $A(x)$ as follows:

$$A : \mathbb{R}^n \to \mathbb{R}^{n\times n} \ , \ A(x)^T = [A_1^T A_1 x, A_2^T A_2 x, \dots, A_n^T A_n x]$$

Then, we have

$$Z'(x) = 2A(x) \ , \ Z(x) = A(x)x$$

Now let $y = Qx$ for an orthogonal matrix Q, define $B_i = A_iQ$, and also extend this definition to the definition of the map $B(x)$ as

$$B(x)^T = [B_1^T B_1 x, \dots, B_n^T B_n x].$$

Then, we have $A(y)y = B(x)x$, but $A(y) \neq B(x)$. However, we note that every row of $B(x)$ is obtained acting on rows of $A(y)$ with the orthogonal matrix Q. So, we naturally have $\det(B(x)) = \det(A(y))$. Now note that rows and columns of A_i are centered Gaussian random vectors, therefore due rotational invariance of Gaussians the distribution of A_i does not change under the action of Q. This means the following:

$$\mathbb{E}\left(|det(Z'(x))| \mid Z(x) = (1, 1, \dots, 1)\right) = \mathbb{E}\left(|det(Z'(y))| \mid Z(y) = (1, 1, \dots, 1)\right)$$

Also note that the density formula Equation (4) is invariant under the action of Q as well. This means using an arbitrary choice $x = e_0$ we can simplify Equation (3) as follows:

$$|S^{n-1}| \int_0^\infty \mathbb{E}\left(|det(Z'(te_1))| \mid Z(te_1) = (1, \dots, 1)\right) \ p_{Z(te_1)}(1, \dots, 1) t^{n-1} \, dt \tag{5}$$

where $|S^{n-1}|$ denotes the volume of the sphere S^{n-1}. Note that

$$Z(te_1) = (1, \dots, 1) \Leftrightarrow Z(e_1) = (\frac{1}{t^2}, \dots, \frac{1}{t^2}) = \left((A_iA_i^T)_{11}\right)_i, \text{ and } Z'(te_1) = tZ'(e_1).$$

Therefore,

$$\mathbb{E}\left(|det(Z'(te_1))| \mid Z(te_1) = (1, \dots, 1)\right) = t^n \, \mathbb{E}\left(|det(Z'(e_1))| \mid Z(e_1) = (\frac{1}{t^2}, \dots, \frac{1}{t^2})\right)$$

Let the columns of the matrices A_i be a_{ij} for $j = 1, 2, \dots, n$. The assumption $Z(e_1) = (\frac{1}{t}, \dots, \frac{1}{t^2})$ is equivalent to $\|a_{i1}\| = \frac{1}{t}$ for all i. Moreover, we have

$$Z'(e_1) = 2 \begin{bmatrix} \langle a_{11}, a_{11}\rangle & \langle a_{11}, a_{12}\rangle & \dots & \langle a_{11}, a_{1n}\rangle \\ \vdots & & & \vdots \\ \langle a_{n1}, a_{n1}\rangle & \langle a_{n1}, a_{n2}\rangle & \dots & \langle a_{n1}, a_{nn}\rangle \end{bmatrix}$$

That is, the assumption $Z(e_1) = (\frac{1}{t}, \dots, \frac{1}{t^2})$ vs $Z(e_1) = (1, \dots, 1)$ scales the determinant with $\frac{1}{t^2} \cdot \frac{1}{t^{n-1}} = \frac{1}{t^{n+1}}$. This yields,

$$\mathbb{E}\left(|det(Z'(te_1))| \mid Z(te_1) = (1, \dots, 1)\right) = t^{-1}\mathbb{E}\left(|det(Z'(e_1))| \mid Z(e_1) = (1, \dots, 1)\right).$$

So, we can further simplify Equation (5) as

$$|S^{n-1}| \, \mathbb{E}\left(|det(Z'(e_1))| \mid Z(e_1) = (1, \dots, 1)\right) \int_0^\infty t^{n-2} p_{Z(te_1)}(1, \dots, 1) \, dt \tag{6}$$

The following proposition deserves a proof. To be done soon

Proposition 3.1.

$$|S^{n-1}|\ \mathbb{E}\left(|det(Z'(e_1))|\ |\ Z(e_1)=(1,\dots,1)\right) \sim c\ \pi^{\frac{n-1}{2}} 2^{n+1} e^{\frac{n}{2}} \sigma^{n-1}\sqrt{n}$$

Proof. (Idea for a proof) Note that assuming $Z(e_1)=(1,1,\dots,1)$ corresponds to that the variance of every entry in the first vector of every matrix A_i is $\frac{1}{n}$. This also means the first column of $Z'(e_1)$ is $(2,2,\dots,2)$. Let Π denote the projection on the orthogonal complement of the line spanned by the vector $(1,1,\dots,1)$. Let $z_2,z_3,\dots,z_n$ denote the second, third, and so on up to n'th column of $Z'(e_1)$. We observe that

$$\mathbb{E}\left(|det(Z'(e_1))|\ |\ Z(e_1)=(1,\dots,1)\right) = \sqrt{n}\mathbb{E}|\det([\Pi(z_2),\Pi(z_n),\dots,\Pi(z_n)])|.$$

Now note that the variance of every entry in the vectors $\Pi(z_2)$ is σ^2. and the random matrix $[\Pi(z_2),\Pi(z_n),\dots,\Pi(z_n)]$ obeys the assumptions of the main result in `https://arxiv.org/pdf/1112.0752.pdf`. So, we should have

$$\mathbb{E}\left(|det(Z'(e_1))|\ |\ Z(e_1)=(1,\dots,1)\right) = c_1 2^n \sigma^{n-1} e^{\frac{n\ln n}{2}}$$

Note that

$$|S^{n-1}| = \frac{2\pi^{\frac{n-1}{2}}}{\Gamma(\frac{n}{2})} \sim 2\pi^{\frac{n-1}{2}} e^{\frac{-(n-1)\ln n + n}{2}}$$

where $c_2<1$. ■

$$\mathbb{E}\left(N_u(Z,\mathbb{R}^n)\right) \sim c\ \pi^{\frac{n-1}{2}} 2^{n+1} e^{\frac{n}{2}} \sigma^{n-1}\sqrt{n} \int_0^\infty t^{n-2} p_{Z(te_1)}(1,\dots,1)\ dt \tag{7}$$

$$p_{Z(te_0)}(1,\dots,1) = \left(2^{\frac{n}{2}}\Gamma(\frac{n}{2})\right)^{-n} (\sigma t)^{-n^2} e^{-\frac{n}{2\sigma^2 t^2}}$$

$$\sigma^{n-1}\int_0^\infty t^{n-2} p_{Z(te_0)}(1,\dots,1)\ dt = \left(2^{\frac{n}{2}}\Gamma(\frac{n}{2})\right)^{-n} \int_0^\infty \sigma^{-1} t^{-2} (\sigma t)^{-n^2+n} e^{-\frac{n}{2\sigma^2 t^2}}\ dt$$

For $u=\frac{1}{\sigma t},\ du = -\frac{1}{\sigma t^2}\ dt$ we have

$$\int_0^\infty \sigma^{-1} t^{-2} (\sigma t)^{-n^2+n} e^{-\frac{n}{2\sigma^2 t^2}}\ dt = \int_0^\infty u^{n^2-n} e^{-\frac{nu^2}{2}}\ du$$

Notice we are having a Gaussian moment here, that is, for $X\sim\mathcal{N}(0,\frac{1}{n})$ we have

$$\mathbb{E}[|X|^{n^2-n}] = \frac{2\sqrt{n}}{\sqrt{2\pi}}\int_0^\infty u^{n^2-n} e^{-\frac{nu^2}{2}}\ du = (\frac{1}{\sqrt{n}})^{n^2-n}\frac{2^{\frac{n^2-n}{2}}\Gamma(\frac{n^2-n+1}{2})}{\sqrt{\pi}}$$

Hence, we have

$$\left(2^{\frac{n}{2}}\Gamma(\frac{n}{2})\right)^{-n} \int_0^\infty \sigma^{-1} t^{-2} (\sigma t)^{-n^2+n} e^{-\frac{n}{2\sigma^2 t^2}}\ dt = \left(2^{\frac{n}{2}}\Gamma(\frac{n}{2})\right)^{-n} n^{-\frac{n^2-n+1}{2}} 2^{\frac{n^2-n-1}{2}} \Gamma(\frac{n^2-n+1}{2}) \tag{8}$$

This gives the following,

$$\mathbb{E}\left(N_u(Z,\mathbb{R}^n)\right) = c\ \pi^{\frac{n-1}{2}} n^{-\frac{n^2-n}{2}} 2^{\frac{n+1}{2}} e^{\frac{n}{2}} \Gamma(\frac{n}{2})^{-n}\Gamma(\frac{n^2-n+1}{2}) \tag{9}$$

Recall that (Stirling):

$$\Gamma(z) \sim \sqrt{2\pi} e^{-z} z^{z-\frac{1}{2}}$$

This implies,

$$\Gamma(\frac{n}{2})^{-n}\Gamma(\frac{n^2-n+1}{2}) \sim (2\pi)^{\frac{-n+1}{2}} e^{\frac{n-1}{2}} (\frac{n}{2})^{-\frac{n^2-n}{2}} (\frac{n^2-n+1}{2})^{\frac{n^2-n}{2}}$$

$$\Gamma(\frac{n}{2})^{-n}\Gamma(\frac{n^2-n+1}{2}) \sim (2\pi)^{\frac{-n+1}{2}} e^{\frac{n-1}{2}} \left(n-1+\frac{1}{n}\right)^{\frac{n^2-n}{2}}$$

$$\mathbb{E}\left(N_u(Z,\mathbb{R}^n)\right) \sim c\, 2e^{\frac{2n-1}{2}} (1-\frac{1}{n}+\frac{1}{n^2})^{\frac{n^2-n}{2}} \tag{10}$$

Note that

$$(1-\frac{1}{n}+\frac{1}{n^2})^{\frac{n^2-n}{2}} \sim \left((1-\frac{1}{n})^n\right)^{\frac{n-1}{2}} \sim e^{-\frac{n-1}{2}}$$

So, we have

$$\mathbb{E}\left(N_u(Z,\mathbb{R}^n)\right) = 2\, c_0\, e^{\frac{n}{2}}$$

where c_0 is a universal constant.

Theorem 3.2. *Let $A_1, A_2, \ldots, A_n$ be random Gaussian matrices with independent entries of standard deviation $\sigma > 0$. Then, the expected number of solutions to the following system of equations*

$$\|A_1 x\|^2 = \|A_2 x\|^2 = \ldots = \|A_n x\|^2 = 1$$

is

$$c_1 e^{\frac{n+2}{2}}.$$

Julia Lindberg performed an experiment as follows. She creates samples of random $n \times n$ Gaussian matrices $A_1, \ldots, A_n$ then does the following: Computes $\|A_i^T A_i\|$ for all $i = 1, 2, \ldots, n$ and then creates the matrices

$$\frac{A_i^T A_i}{\|A_i^T A_i\|} \text{ for } i = 1, 2, \ldots, n.$$

Then she solves the corresponding equations and counts the number of real zeros. Then she computes the average number of real zeros for $2 < n \leq 10$. This data and model fitting Jupyter notebook is included in our repo. If we look at the fit in Jupyter notebook based on Julia's data, it says the fit is

$$2.0009e^{0.352n} = 2.0009(1.4219)^n \sim (2.02)^{\frac{n+2}{2}}$$

Note that this experiment is different than our model of randomness in this note.

4. Integral formula for random ellipsoids with prescribed centers

We use the notation from Section 2.2: We have $A_i = B_i + \delta X_i$ where X_i are Gaussian random matrices with indepedent entries centered at zero and variance one, $1 > \delta > 0$ is our perturbation parameter, and B_i are deterministic matrices that satisfy the following:

$$\|B_i\|_F = 1 \text{ for } i = 1, 2, \ldots, n, \text{ and } B_1^T B_1 + B_2^T B_2 + \ldots + B_n^T B_n = \mathbb{I}.$$

In this setting, we denote the average number of solutions to the system of equations

$$\|A_1 x\|^2 = \|A_2 x\|^2 = \ldots = \|A_n x\|^2 = 1$$

with $\mathbb{E}N(\mathcal{B},\delta)$ where $\mathcal{B}=(B_1,B_2,\ldots,B_n)$ for the center of our random perturbation. Following the discussion Section 2.2 we have

$$\mathbb{E}N(\mathcal{B},\delta)=\int_{\mathbb{R}^n}\int_{K(x)}|\det Z'(x)|\,d\mu(X_1,X_2,\ldots,X_n)\,dx.$$

We noted that the assumption on B_i imply that $\mu(K(x))$ is very small unless $\|x\|\sim\frac{\sqrt{n}}{\sqrt{1+\delta^2 n}}$. We should also note that for $A_i=B_i+\delta X_i$ to serve as decent proxy to B_i we must have

$$\delta n=\mathbb{E}\|\delta X_i\|_F<\|B_i\|\Rightarrow\delta<\frac{1}{n}.$$

So, we restrict $0<\delta\leq\frac{1}{n}$ and in this parameter regime we must have $\|x\|\sim\sqrt{n}$ in order for $\mu(K(x))$ to be non-negligible. We will add an $\varepsilon>\frac{4}{\sqrt{n}}$ tolerance, and we will aim to sample vectors uniformly from the annulus

$$(1-\varepsilon)\sqrt{n}<\|x\|<(1+\varepsilon)\sqrt{n}$$

Here we recall two facts about the Gaussian measure: (1) Gaussian measure is rotationally invariant, (2) For standard Gaussian vector $x\in\mathbb{R}^n$, we have

$$\mathbb{P}\{(1-\varepsilon)\sqrt{n}\leq\|x\|\leq\frac{\sqrt{n}}{1-\varepsilon}\}\geq 1-2e^{-\frac{\varepsilon^2 n}{4}}$$

Since we fix $\varepsilon>\frac{4}{\sqrt{n}}$ we have

$$\mathbb{P}\{(1-\varepsilon)\sqrt{n}\leq\|x\|\leq\frac{\sqrt{n}}{1-\varepsilon}\}\geq 1-2e^{-4}\geq 0.97$$

We can use a conditional density scaled with $\frac{1}{0.97}$, but we prefer not to bother with this small, we'll directly use standard Gaussians to sample from the annulus $(1-\varepsilon)\sqrt{n}<\|x\|<(1+\varepsilon)\sqrt{n}$. So, our Monte-Carlo approximation to $\mathbb{E}N(\mathcal{B},\delta)$ will go as follows:

(1) Sample $x_1,x_2,\ldots,x_N\in\mathbb{R}^n$ independently from standard Gaussian measure on $\mathbb{R}^n$. Here N will be determined by the user, it will determine the quality of approximation.
(2) For every x_i, sample M elements from $K(x_i)$. This will be done by sampling Gaussian tuples and projecting them on $K(x_i)$. Let's denote the resulting sample with $(X_1^s,X_2^s,\ldots,X_n^s)_{s=1}^M$.
(3) We use $A(x_i,X_1^s,\ldots,X_n^s)$ to denote the following matrix

$$A(x_i,X_1^s,\ldots,X_n^s):=\begin{bmatrix}(B_1+\delta X_1^s)^T(B_1+\delta X_1^s)x_i\\(B_2+\delta X_2^s)^T(B_2+\delta X_2^s)x_i\\\vdots\\(B_n+\delta X_n^s)^T(B_n+\delta X_n^s)x_i\end{bmatrix}$$

and we compute

$$\frac{1}{N}\sum_{i=1}^{N}\sum_{s=1}^{M}|\det A(x_i,X_1^s,\ldots,X_n^s)|$$